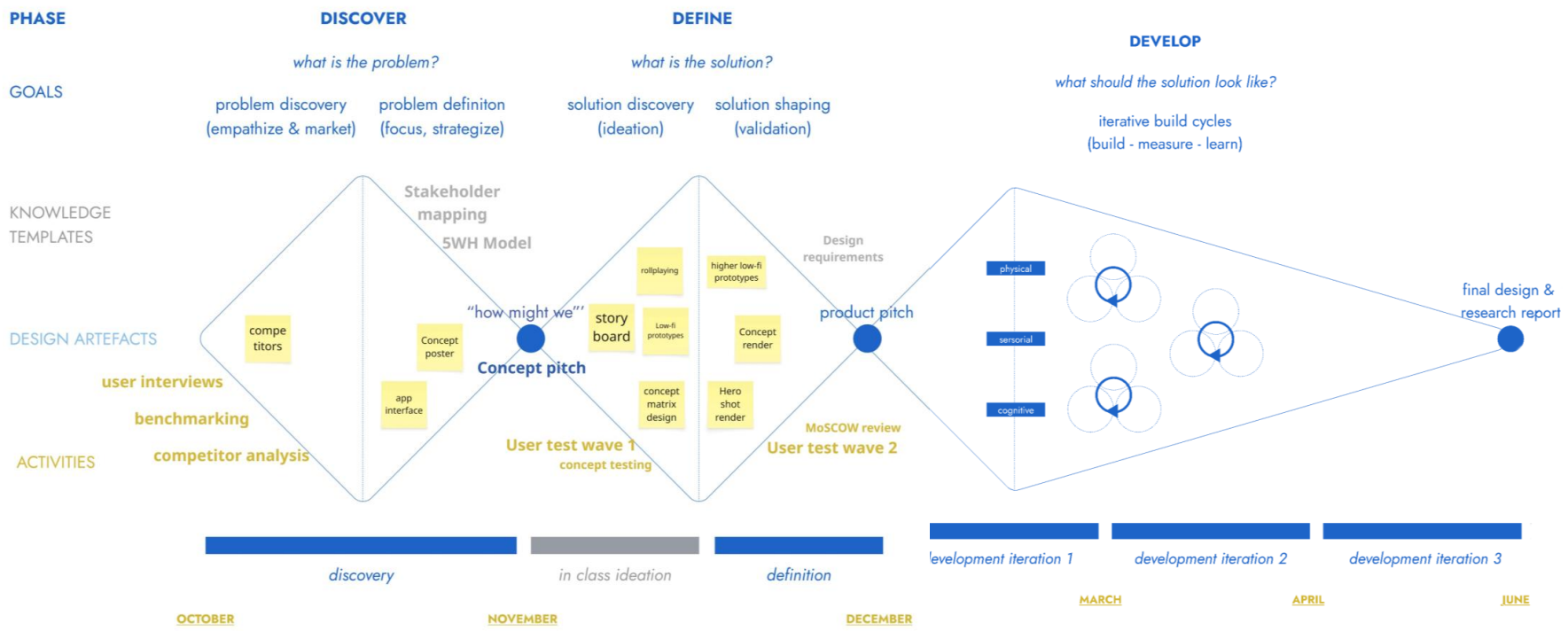
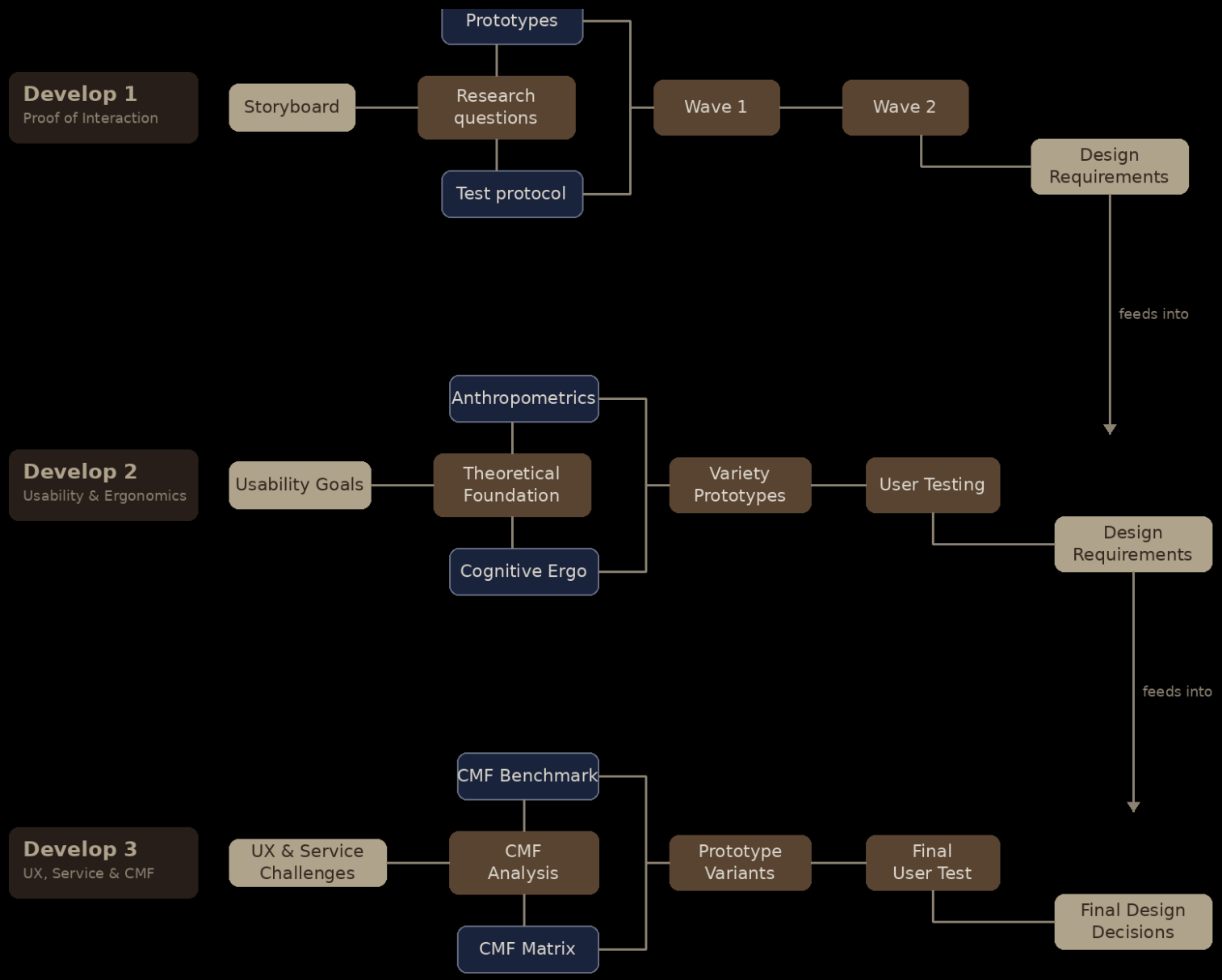
SensePath

Ear free, intuitive navigation in & outdoors


PHASE
DISCOVER
DEFINE
DEVELOP
GOALS
what is the problem?
what is the solution?
what should the solution look like?
problem discovery (empathize & market)
problem definiton (focus, strategize)
solution discovery (ideation)
solution shaping (validation)
iterative build cycles (build - measure - learn)
KNOWLEDGE TEMPLATES
Stakeholder mapping
5WH Model
Design requirements
DESIGN ARTEFACTS
compe titors
Concept poster
app interface
"how might we"'
Concept pitch
story board
rollplaying
Low-fi prototypes
concept matrix design
higher low-fi prototypes
Concept render
Hero shot render
product pitch
physical
sersorial
cognitive
final design & research report
user interviews
benchmarking
competitor analysis
ACTIVITIES
User test wave 1
concept testing
MoSCOW review
User test wave 2
discovery
in class ideation
definition
development iteration 1
development iteration 2
development iteration 3
OCTOBER
NOVEMBER
DECEMBER
MARCH
APRIL
JUNE

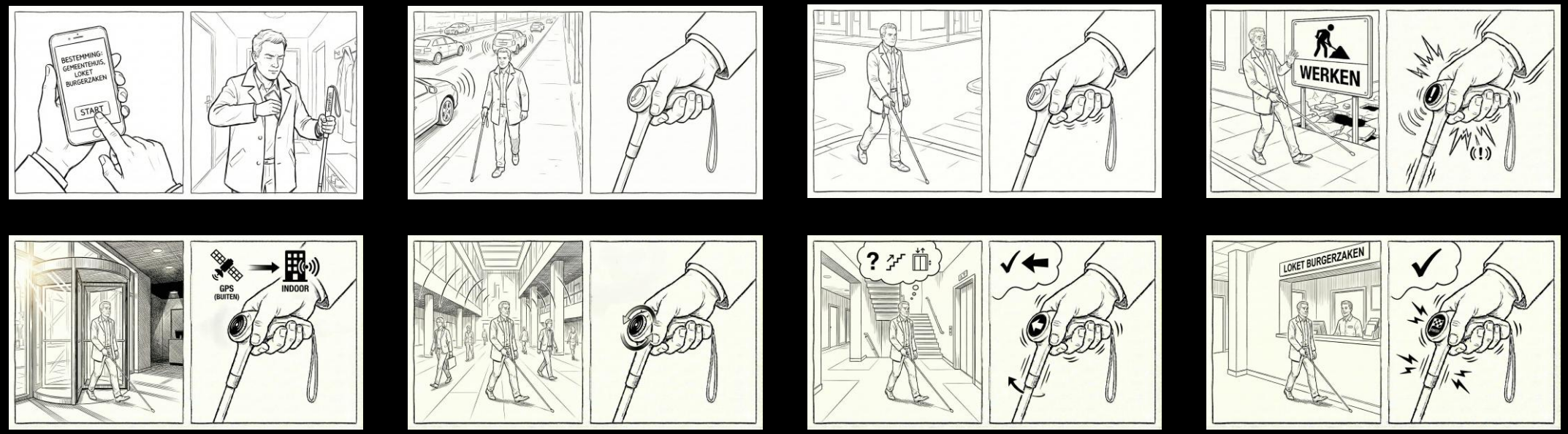
Develop

Methodology



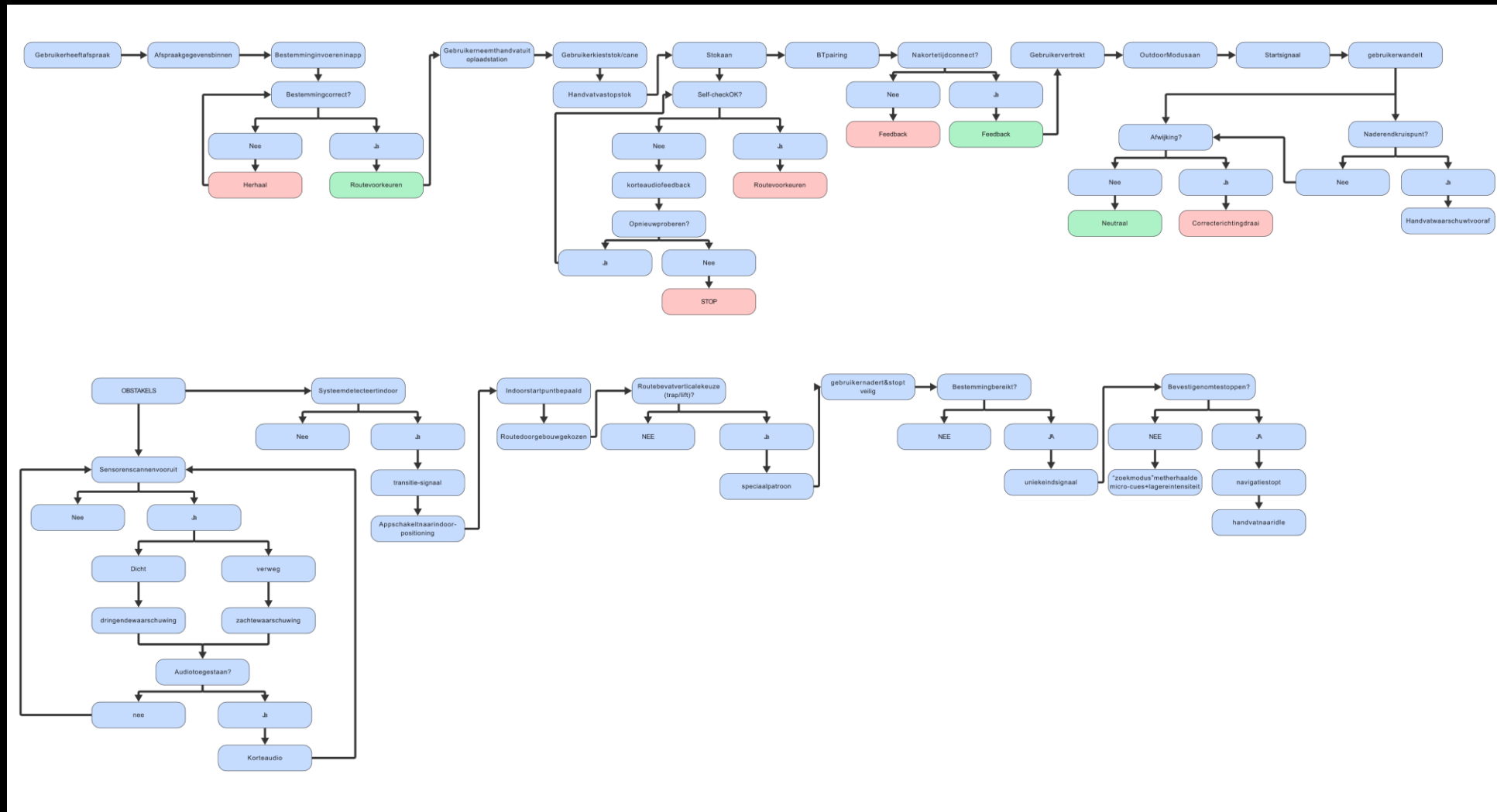
Reassess

Storyboard



Script: Claude Code · Image direction: Chat GPT · Generation: NanoBanana

User-flow

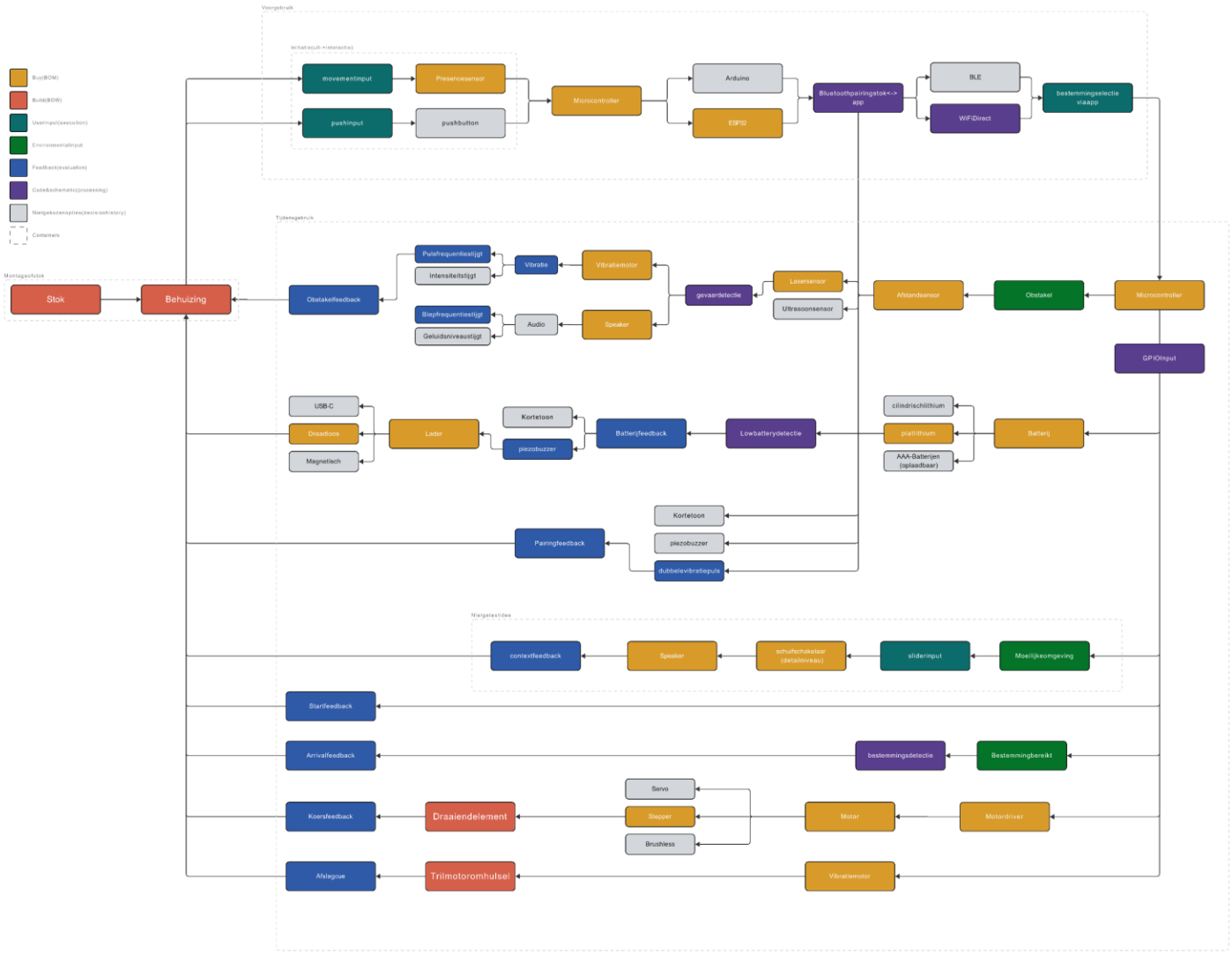

Gebruikerheeftafspraak
Afspraakgegevensbinnen
Bestemminginvoereninapp
Bestemmingcorrect?
Nee
Herhaal
Ja
Routevoorkeuren
Gebruikerneemthandvatuit oplaadstation
Gebruikerkieststok/cane
Handvatvastopstok
Stokaan
Self-checkOK?
Nee
korteaudiofeedback
Opnieuwproberen?
Ja
Nee
STOP
Ja
Routevoorkeuren
BTpairing
Nakortetijdconnect?
Nee
Feedback
Ja
Feedback
Gebruikervertrekt
OutdoorModusaan
Startsignaal
gebruikerwandelt
Afwijking?
Nee
Neutraal
Ja
Correcterichtingdraai
Naderendkruispunt?
Nee
Ja
Handvatwaarschuwtvooraf
OBSTAKELS
Sensorenscannenvooruit
Nee
Ja
Dicht
verweg
dringendewaarschuwing
zachtewaarschuwing
Audiotoegestaan?
nee
Ja
Korteaudio
Systeemdetecteertindoor
Nee
Ja
transitie-signaal
Appschakeltnaarindoor-positioning
Indoorstartpuntbepaald
Routedoorgebouwgekozen
Routebevatverticalekeuze (trap/lift)?
NEE
Ja
speciaalpatroon
gebruikernadert&stopt veilig
Bestemmingbereikt?
NEE
JA
uniekeindsignaal
Bevestigenomtestoppen?
NEE
"zoekmodus"metherhaalde micro-cues+lagereintensiteit
JA
navigatiestopt
handvatnaaridle

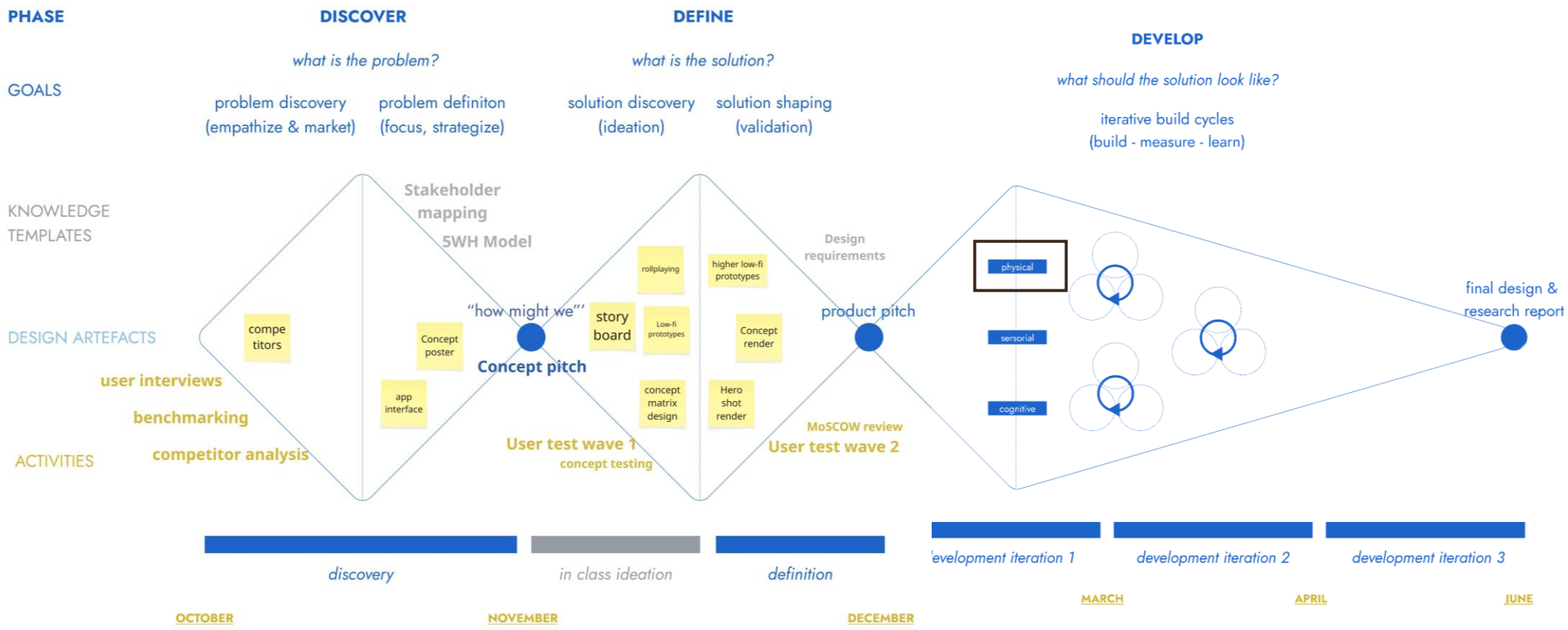
Matrix

- Keuze prototype
- Keuze finaal
- Keuze beide

	Optie 1	Optie 2	Optie 3	Optie 4
Activatie-input	presence sensor	druk knop	schuifknop	touch knop
Microcontroller	ESP32	ESP32-S3	nRF52840	Arduino Nano ESP32
Communicatie protocol	WiFI Direct	BLE	NFC + BLE	/
Afstandsensor	Lasersensor	Ultrasoon sensor	/	/
Batterij	Lithium - cilindrisch	Lithium - plat	AAA - oplaadbaar	/
Oplaad module	USB-C	Draadloos	Magnetisch	/
Kompas motor	Servo	Stepper	Brushless	/
Afslag cue	trilmotor	solenoide	voice coil actuator	lineaire actuator
Speaker	geen	micro speaker	mini dynamische speaker	bone conduction transducer
Stokverbinding	Schroef	wrijving	klemming	twist-lock
Handinterface	groeven	verhoogde feedback-eilandjes	adaptieve ergonomische grip	feedback relief zones
Route selectie	preset locaties	spraak	dropdown	typen
Kompasrichting	tov telefoon orientatie	tov noorden	tov IMU clip	tov stok

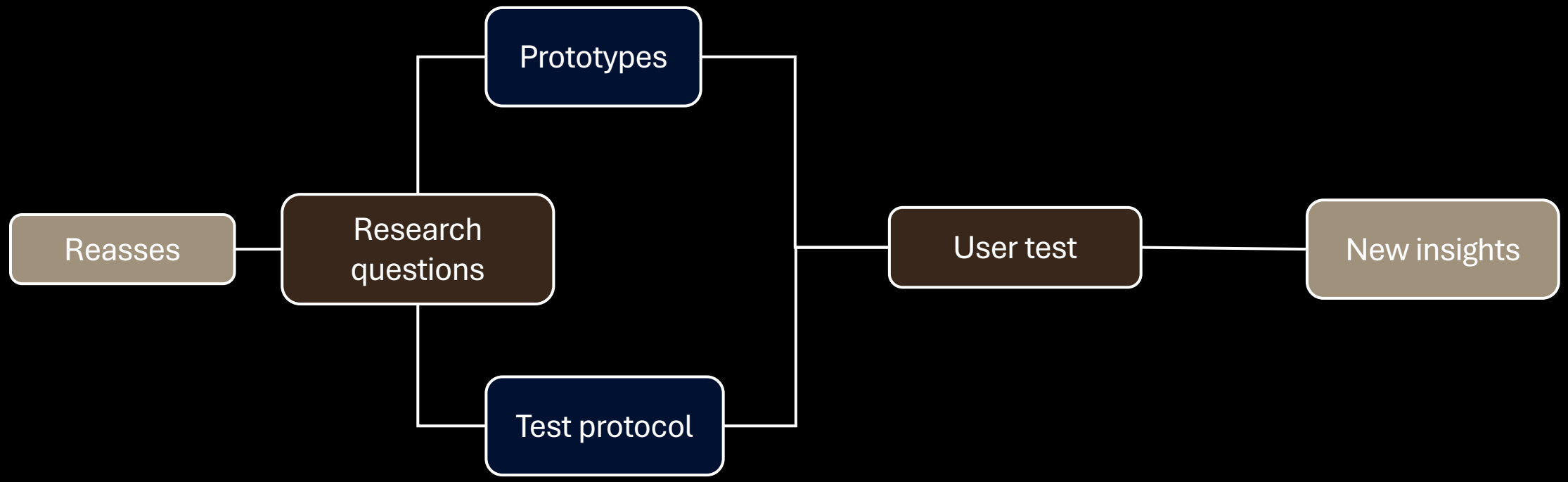
HPI


Buy(BOM)
Build(BOM)
Userinput(execution)
Environmentalinput
Feedback(evaluation)
Code&schematic(processing)
Noteigekozenopties(decisionhistory)
Containers
Montageofstok
Stok
Behuizing
Voorgebruik
Initiatiejuit->interactie)
movementinput
Presencesensor
pushinput
pushbutton
Microcontroller
Arduino
ESP32
Bluetoothpairingstok<->app
BLE
WiFiDirect
bestemmingsselectie viaapp
Tijdensgebruik
Obstakelfeedback
Pulsfrequentiestijgt
Intensiteitstijgt
Vibratie
Vibratiemotor
Biepfrequentiestijgt
Geluidsniveaustijgt
Audio
Speaker
gevaardetectie
Lasersensor
Ultrasoonsensor
Afstandsensor
Obstakel
Microcontroller
GPIOInput
USB-C
Draadloos
Magnetisch
Lader
Kortetoon
piezobuzzer
Batterijfeedback
Lowbatterydetectie
cilindrischlithium
platlithium
AAA-Batterijen (oplaadbaar)
Batterij
Kortetoon
piezobuzzer
dubbelavibratiepuls
Pairingfeedback
Navigatieslide
contextfeedback
Speaker
schuifschakelaar (detailniveau)
sliderinput
Moeilijkeomgeving
Startfeedback
Arrivalfeedback
bestemmingsdetectie
Bestemmingbereikt
Koersfeedback
Draaiendelement
Servo
Stepper
Brushless
Motor
Motordriver
Afslagcue
Trilmotoromhulsel
Vibratiemotor


PHASE
DISCOVER
DEFINE
DEVELOP
GOALS
what is the problem?
what is the solution?
what should the solution look like?
problem discovery (empathize & market)
problem definiton (focus, strategize)
solution discovery (ideation)
solution shaping (validation)
iterative build cycles (build - measure - learn)
KNOWLEDGE TEMPLATES
Stakeholder mapping
5WH Model
Design requirements
DESIGN ARTEFACTS
compe titors
Concept poster
app interface
"how might we"
Concept pitch
story board
rollplaying
Low-fi prototypes
concept matrix design
higher low-fi prototypes
Concept render
Hero shot render
product pitch
physical
sersorial
cognitive
final design & research report
user interviews
benchmarking
competitor analysis
ACTIVITIES
User test wave 1
concept testing
MoSCOW review
User test wave 2
discovery
in class ideation
definition
evelopment iteration 1
development iteration 2
development iteration 3
OCTOBER
NOVEMBER
DECEMBER
MARCH
APRIL
JUNE

Develop 1

Methodology



Research Questions

- Can the user correctly and timely identify and execute a turn (left, right, diagonal) via a vibration signal in the handle?
- Can the user correctly and timely identify and execute a turn via continuous compass feedback in the handle?
- How effective is the combination of vibration (event-cue) and compass feedback (continuous course correction) on a combined course?
- How does accuracy (out-of-area proportion) compare to the Tactile Compass benchmark (Liu et al., CHI 2021)?
- What is the subjective experience (workload, trust, learnability) of the user in each condition?

Goal

Validate haptic interaction

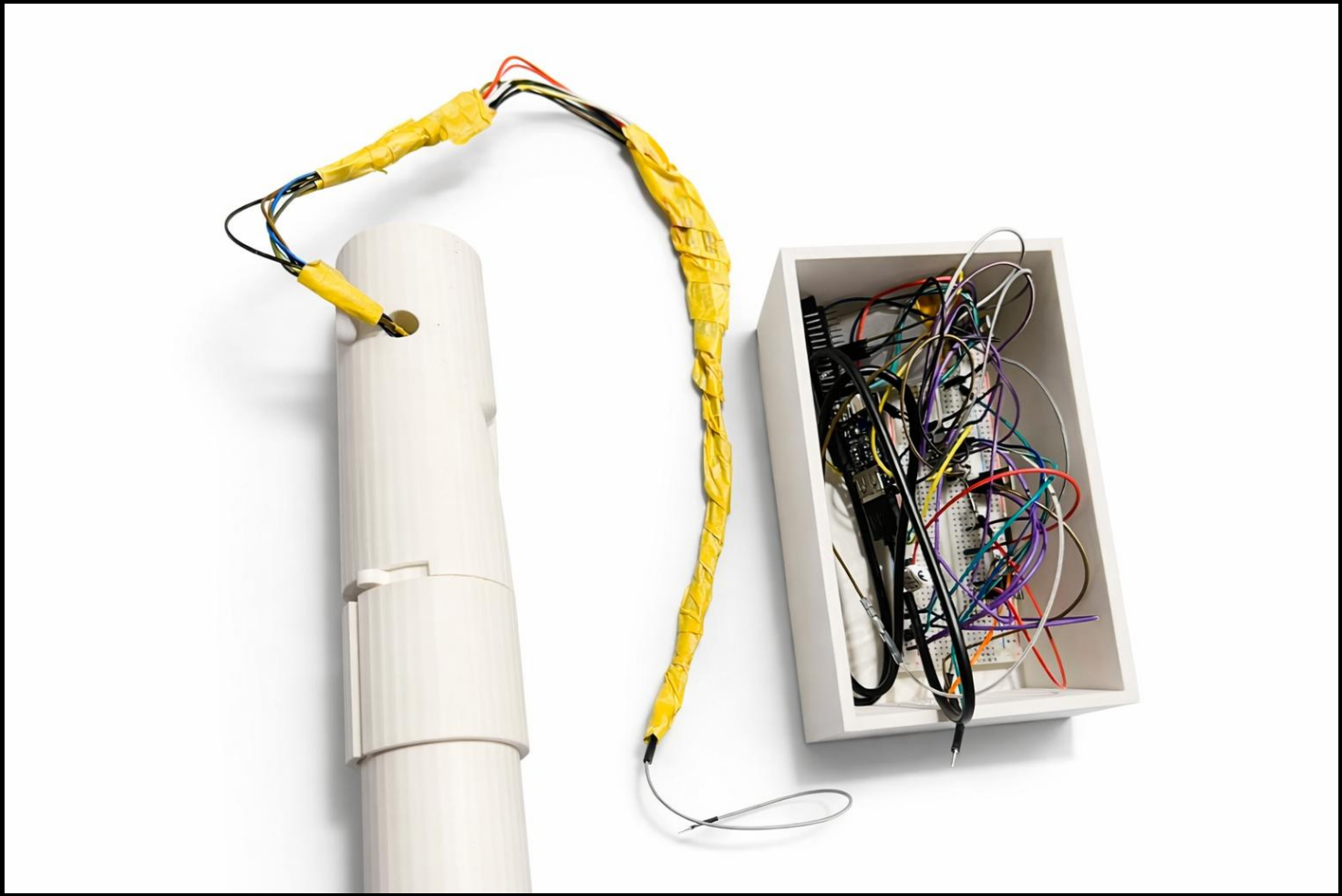
Assess **cognitive load**

Benchmark against literature

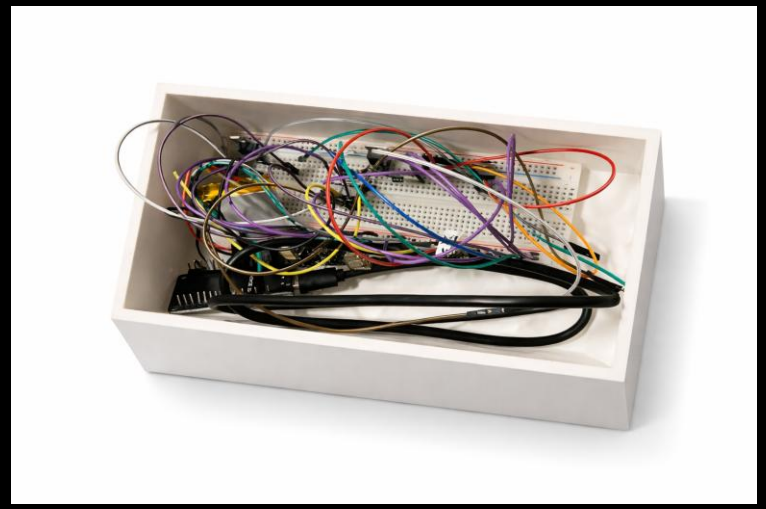
Sample N=5

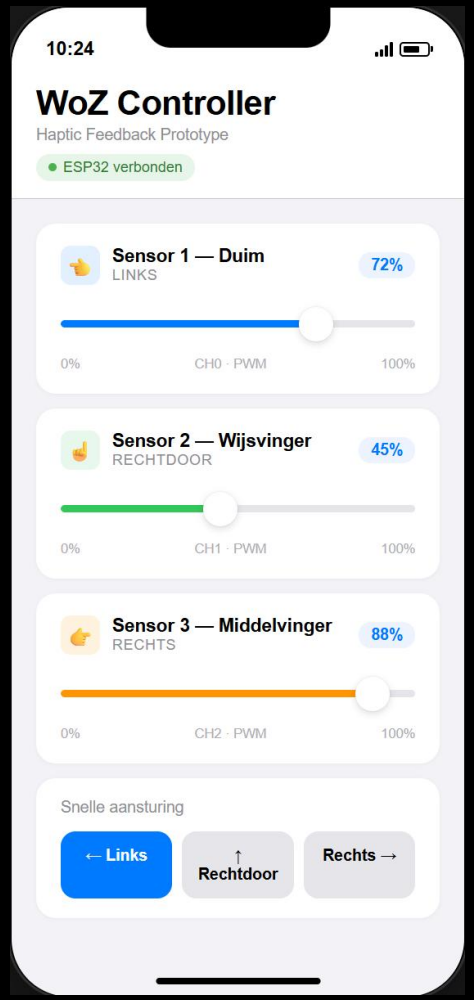
Name	Age	Relevance	Location	Time
Mario	52	Lost vision at age 18	Licht en liefde	04/03/2026
Rory	55	Severely visually impaired	Licht en liefde	04/03/2026
Herman	65	Blind since birth	Licht en liefde	06/03/2026
Milo	20	Sighted (control)	At home	08/03/2026
Milos	20	Sighted (control)	At home	08/03/2026

Prototype



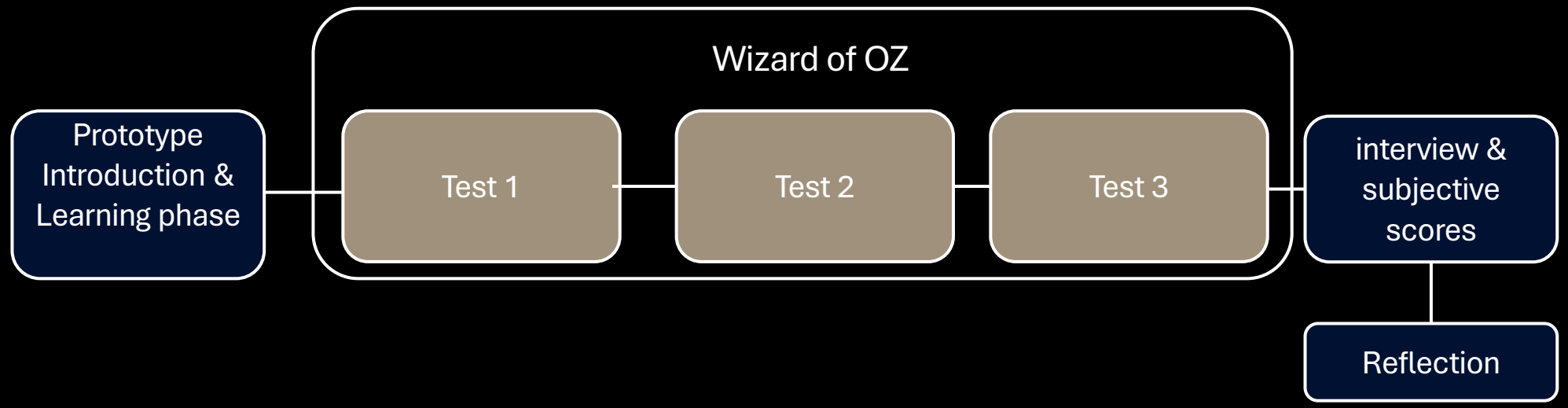
Prototype: vibration





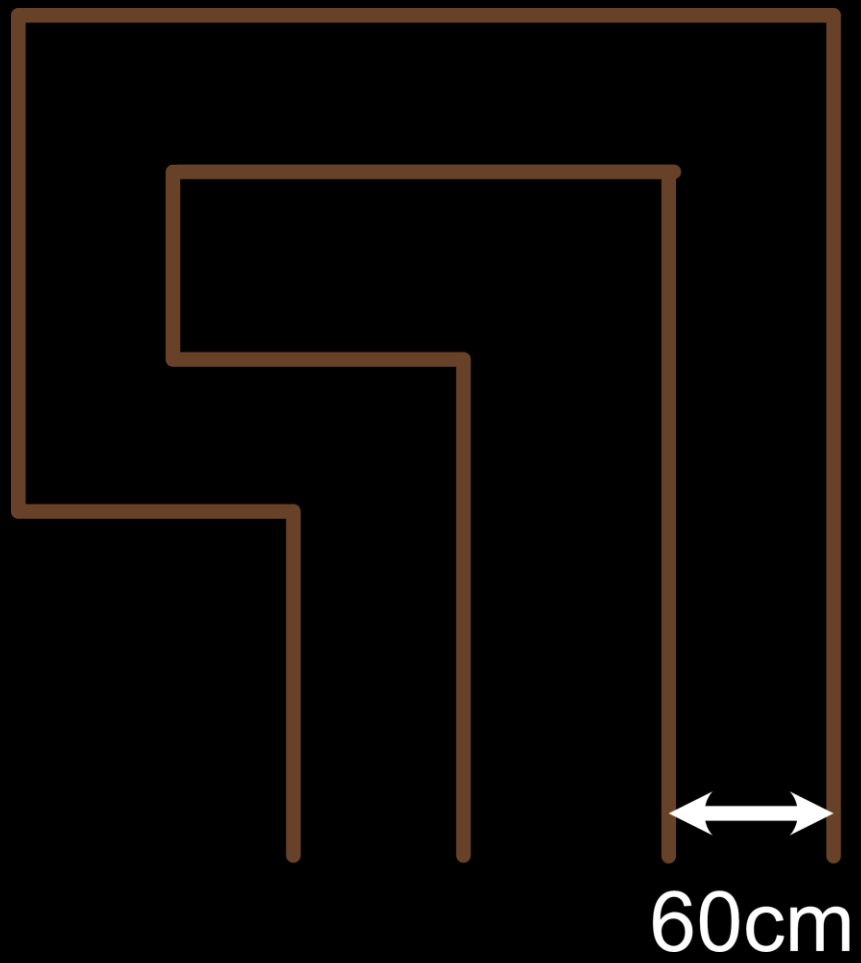
Prototype: Compass

Workflow



Protocol: protocol_sensepath_develop1 (1).docx

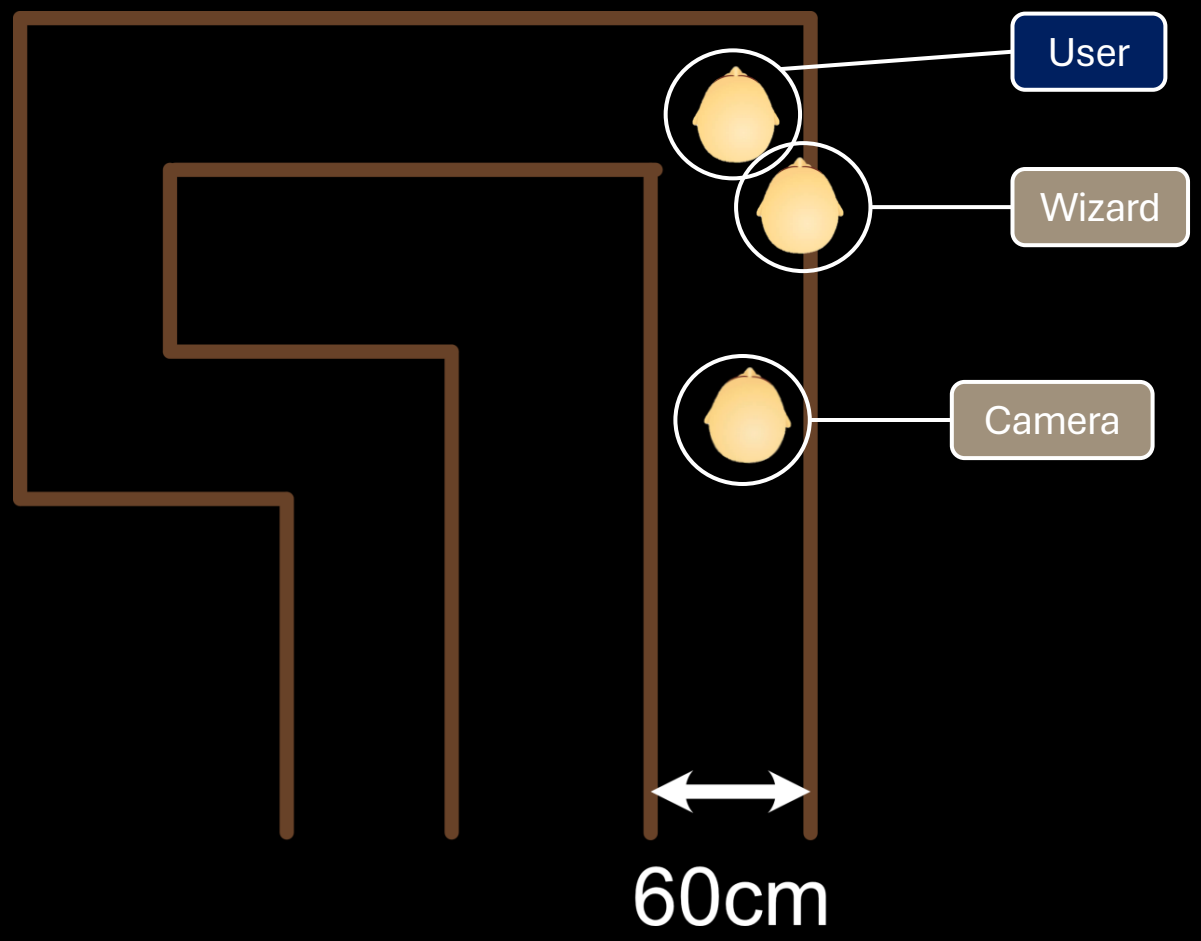
Path



Benchmark comparison

Metric	Tactile Compass(Liu et al.)	SensePath (Develop 1)
Walking zone width	60 cm	
TIZ (% Time in zone) VS SIZ (% Steps in zone)	92,6% (Tactus-Only)	
Apprenticeship	Average. 4 min	
Learnability (1–7)	6,67	
Smoothness (1–7)	6,06	
Cognitive load (1–7)	5,22	
Willingness (1–7)	6,39	
Trust (1–7)	n.v.t.	
Positioning system	OptiTrack (mm-precision)	
Stick used?	No (loose handle)	
Context	Lab (7×13 m)	

Test 1: Turn via compass feedback



- Wizard provides continuous feedback via handle compass

Test 1: Feedback

- **Grip determines interpretation** → Different grip = left/right reversed. Hand orientation is crucial.
- **Palm sensation is poor** → Multiple participants barely or cannot feel the ball.
- **Ergonomics insufficient** → Handle doesn't fit every hand. Ball is mispositioned.
- **Straight walking works** → Good results when walking straight. Turns are harder.
- **Physical reference remains necessary** → "Fully relying on electronics, that can fail.“
- **Winter = problem** → Gloves and thick coats block the sensation.

“

"Ik denk dat het zelfs zou lukken om niet buiten het pad te gaan.“

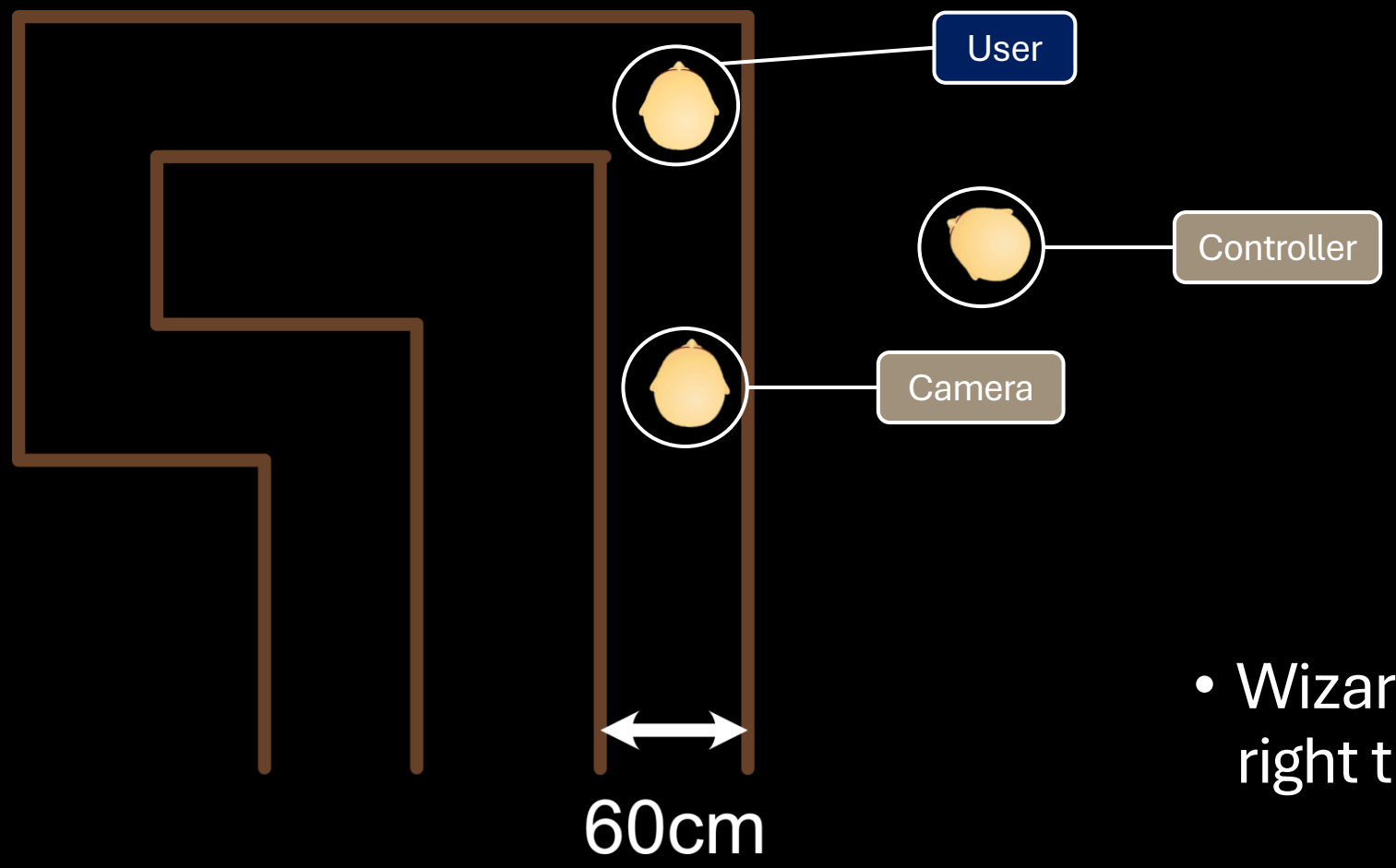
"Dat was eigenlijk redelijk goed eigenlijk."

”

Test 1: Benchmark comparison

Metric	Tactile Compass(Liu et al.)	SensePath (Develop 1)
Walking zone width	60 cm	60cm
TIZ (% Time in zone) VS SIZ (% Steps in zone)	92,6% (Tactus-Only)	83,78%
apprenticeship	Av. 4 min	Av. 4min
Learnability (1–7)	6,67	5,67
Smoothness (1–7)	6,06	5,50
Cognitive load (1–7)	5,22	3,67
Willingness (1–7)	6,39	5,33
Trust (1–7)	n.v.t.	5,33
Positioning system	OptiTrack (mm-precision)	Video+Observation
Stick used?	No (loose handle)	Yes
Context	Lab (7×13 m)	Interior space licht en liefde

Test 2: Turn via vibration signal



- Wizard sends signals at the right time

Test 2: Feedback

- **Fingertips feel best** → "That's where the feelers are.“
- **Turning works well** → Directional signals quickly understood and learned.
- **Staying straight is harder** → Without continuous signal, users drift off course.
- **Confirmation signal needed** → Users want feedback that they're walking straight.
- **Not too many signals** → "Don't start with seven hundred and thirty-eight different signals.“
- **Sensor position problematic** → Left fits well = right fits poorly, and vice versa. Hand size matters.
- **Middle sensor feels weaker** → Poorly perceptible for multiple users.

"Die trilmotoren, dat ging wel goed”

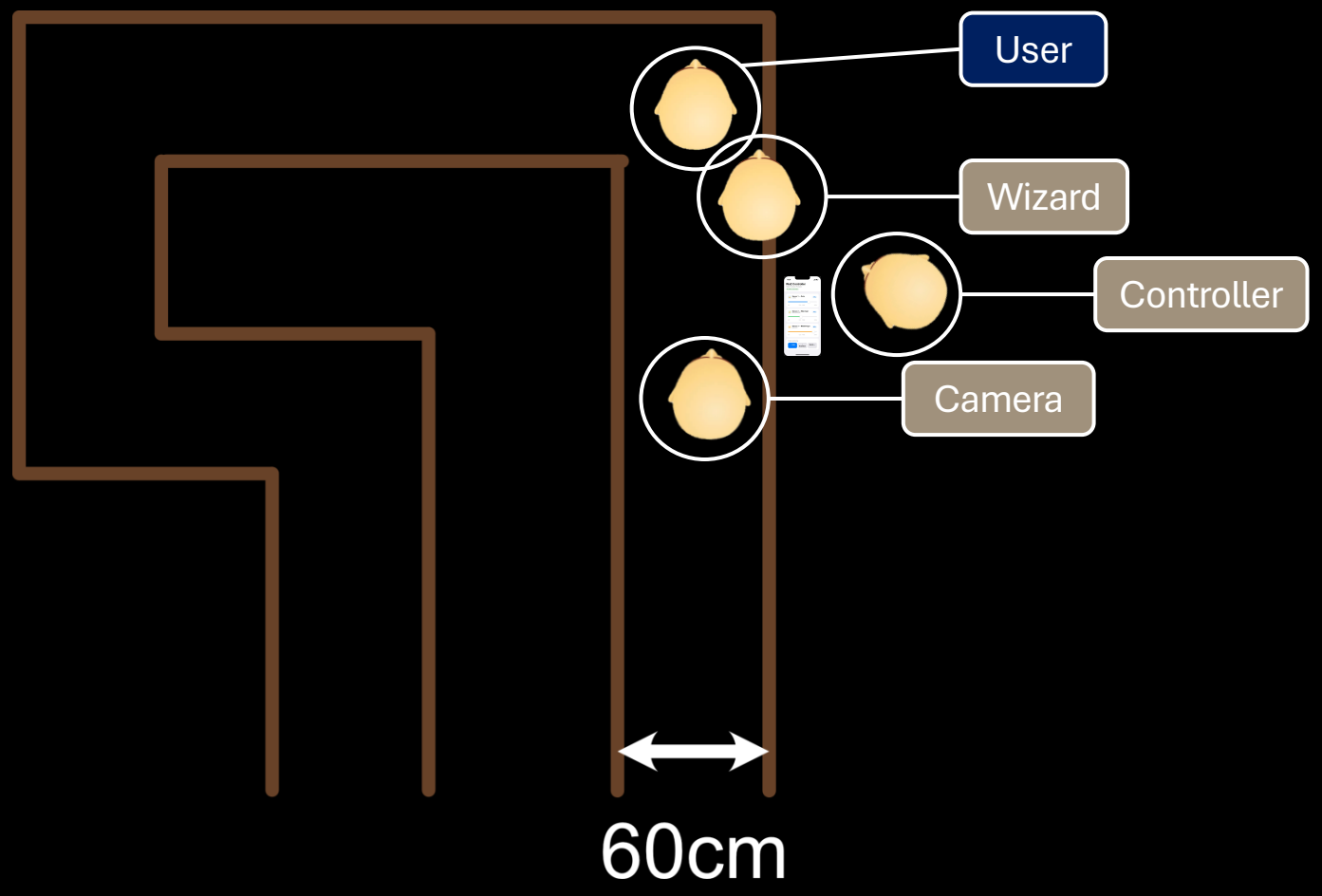
"Het voelde beter, minder concentreren dan het balletje.“

"De kortslag was wel goed."

Test 2: Benchmark comparison

Metric	Tactile Compass(Liu et al.)	SensePath (Develop 1)
Walking zone width	60 cm	60cm
TIZ (% Time in zone) VS SIZ (% Steps in zone)	92,6% (Tactus-Only)	79,85%
apprenticeship	Av. 4 min	Av. 4min
Learnability (1–7)	6,67	6,50
Smoothness (1–7)	6,06	6,33
Cognitive load (1–7)	5,22	4,67
Willingness (1–7)	6,39	6,33
Trust (1–7)	n.v.t.	6
Positioning system	OptiTrack (mm-precision)	Video+Observation
Stick used?	No (loose handle)	Yes
Context	Lab (7×13 m)	Interior space licht en liefde

Test 3: Combined course



Test 3: Feedback

- **Vibrations interfere with compass** → Reading two systems simultaneously on one hand is difficult.
- **Ergonomics is the bottleneck** → Current prototype doesn't allow a fair evaluation.
- **Concept has potential** → Vibration as announcement + compass for course correction is logical.
- **Higher cognitive load** → Scores drop compared to individual tests.
- **Relocate the ball** → Suggestion: to the knuckles or heel of the hand instead of the palm.
- **Audio as back-up** → Speaker/bluetooth for when vibrations are not perceptible.

"Ik ben wel onder indruk van deze keer."

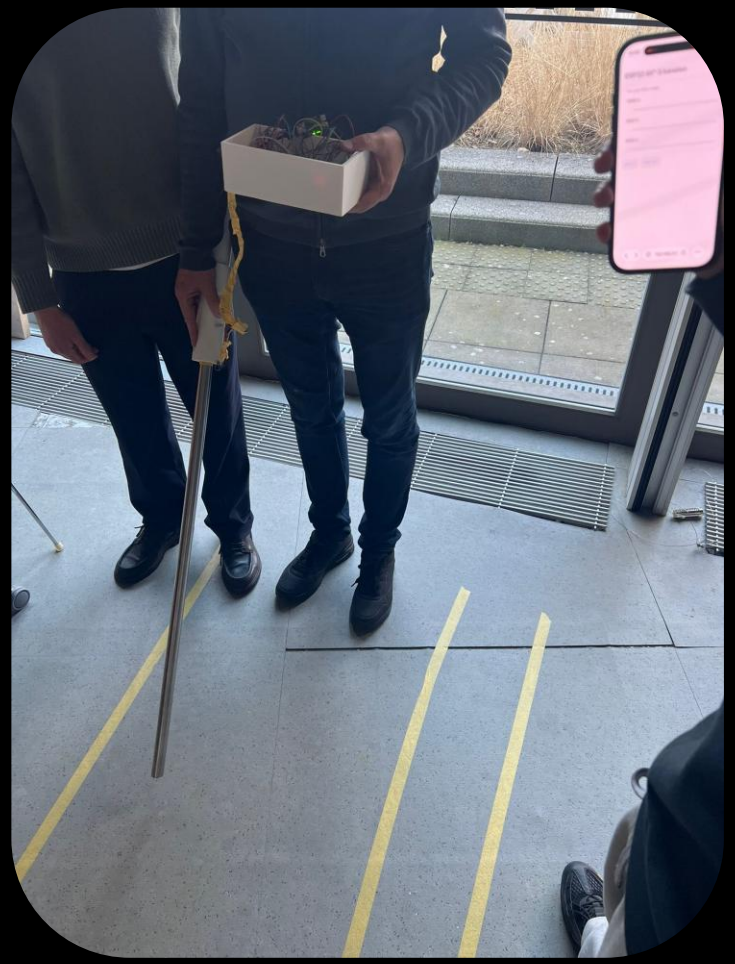
"Het was beter dan ik had verwacht."

"Laatst moest wat bijsturen. Da's de bedoeling van het systeem."

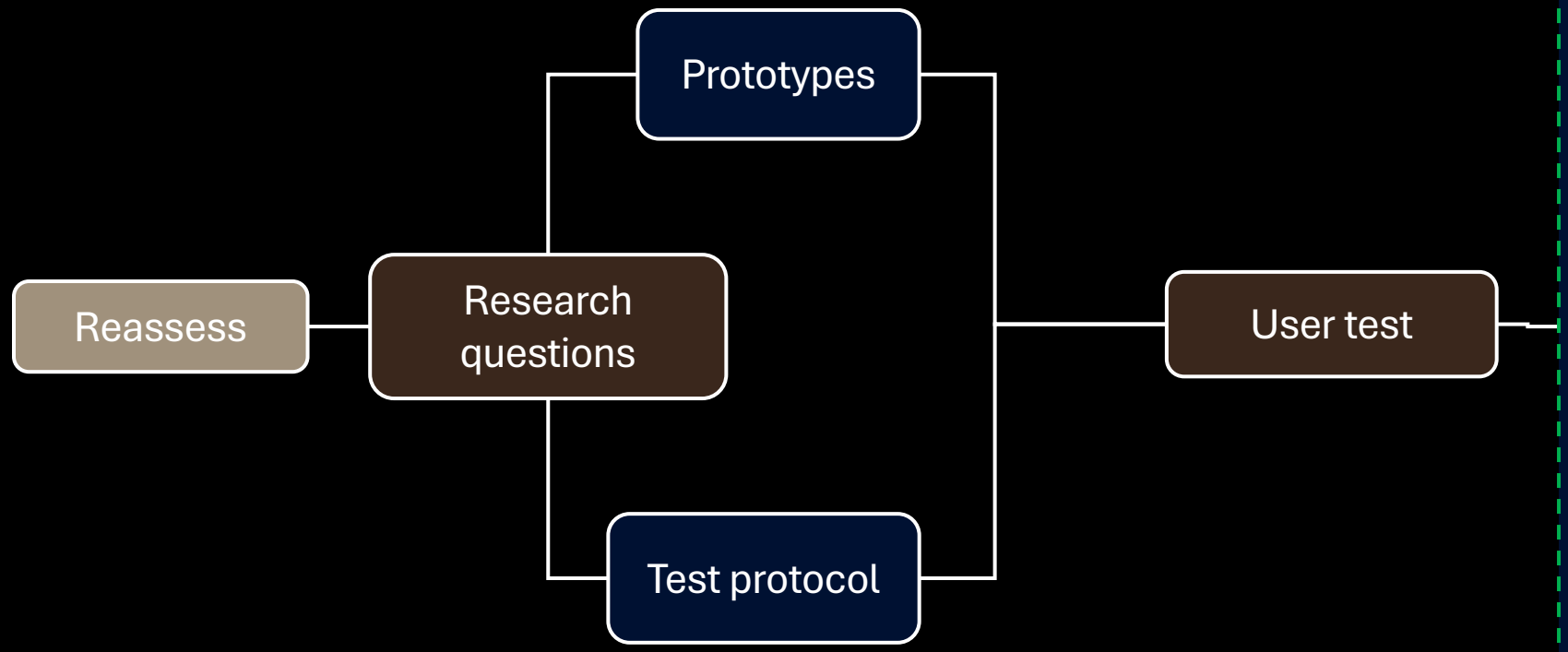
Test 3: Benchmark comparison

Metric	Tactile Compass(Liu et al.)	SensePath (Develop 1)
Walking zone width	60 cm	60cm
TIZ (% Time in zone) VS SIZ (% Steps in zone)	92,6% (Tactus-Only)	97,52%
Apprenticeship	Av. 4 min	Av. 4min
Learnability (1–7)	6,67	6,00
Smoothness (1–7)	6,06	4,67
Cognitive load (1–7)	5,22	3
Willingness (1–7)	6,39	4,33
Trust (1–7)	n.v.t.	3
Positioning system	OptiTrack (mm-precision)	Video+Observation
Stick used?	No (loose handle)	Yes
Context	Lab (7×13 m)	Interior space licht en liefde

Vibe check



Develop 1 conclusions

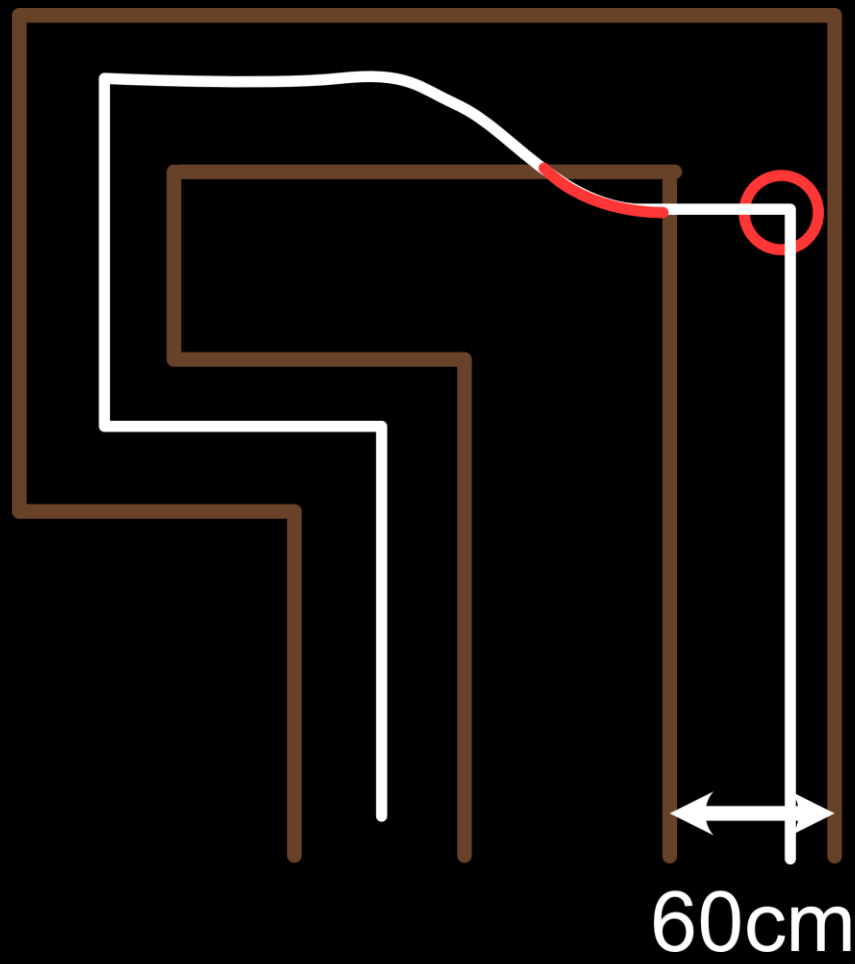


- Vibration is more intuitive than compass for discrete navigation events.
- Compass provides superior path-keeping but suffers from perceptual ambiguity
- The combined condition scored highest on accuracy (97.52% SIZ) but lowest on subjective experience.
- Ergonomics is not a side issue — it is the primary confound in the data.
- SensePath holds up against the Tactile Compass benchmark under harder conditions.

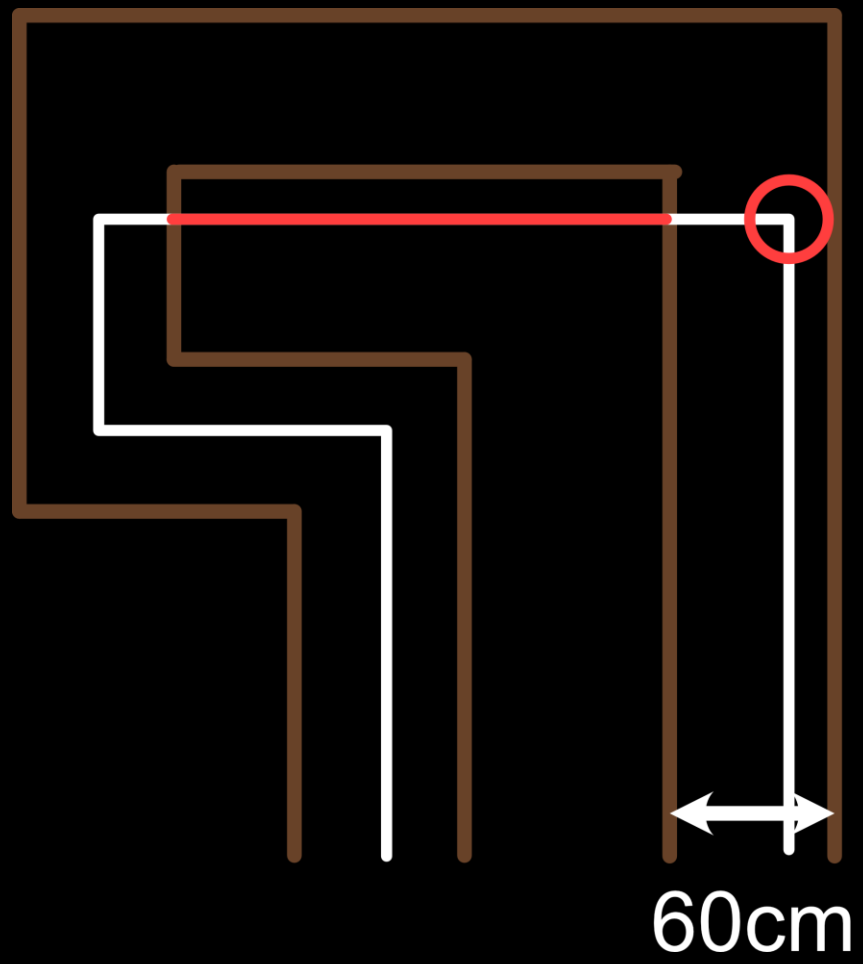
Critical reflection

Path

Test 1

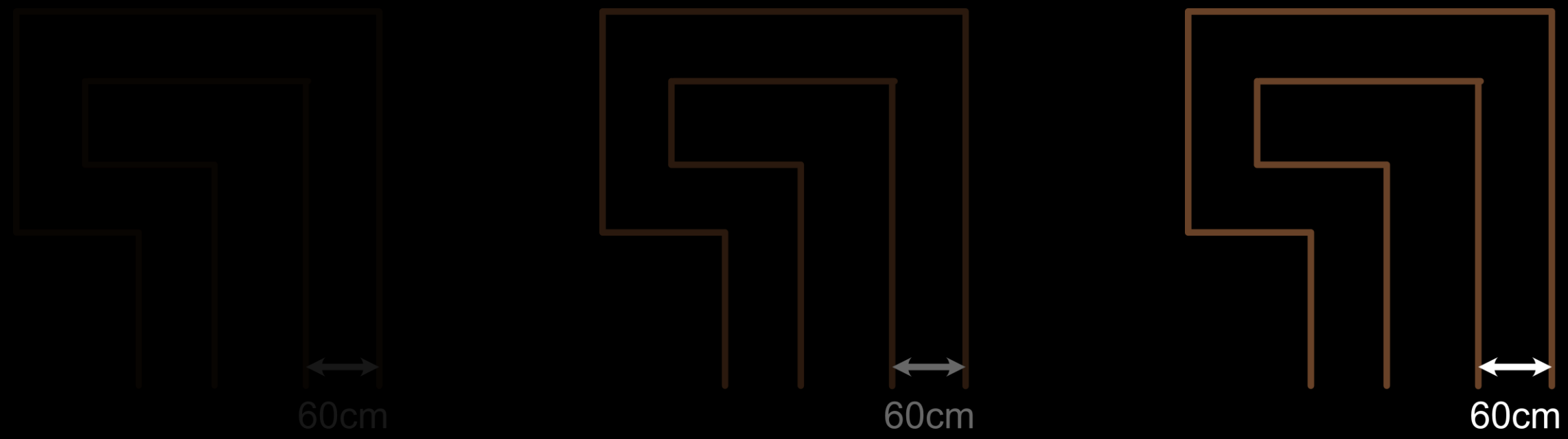


Test 2



Faster correction

Path Learning



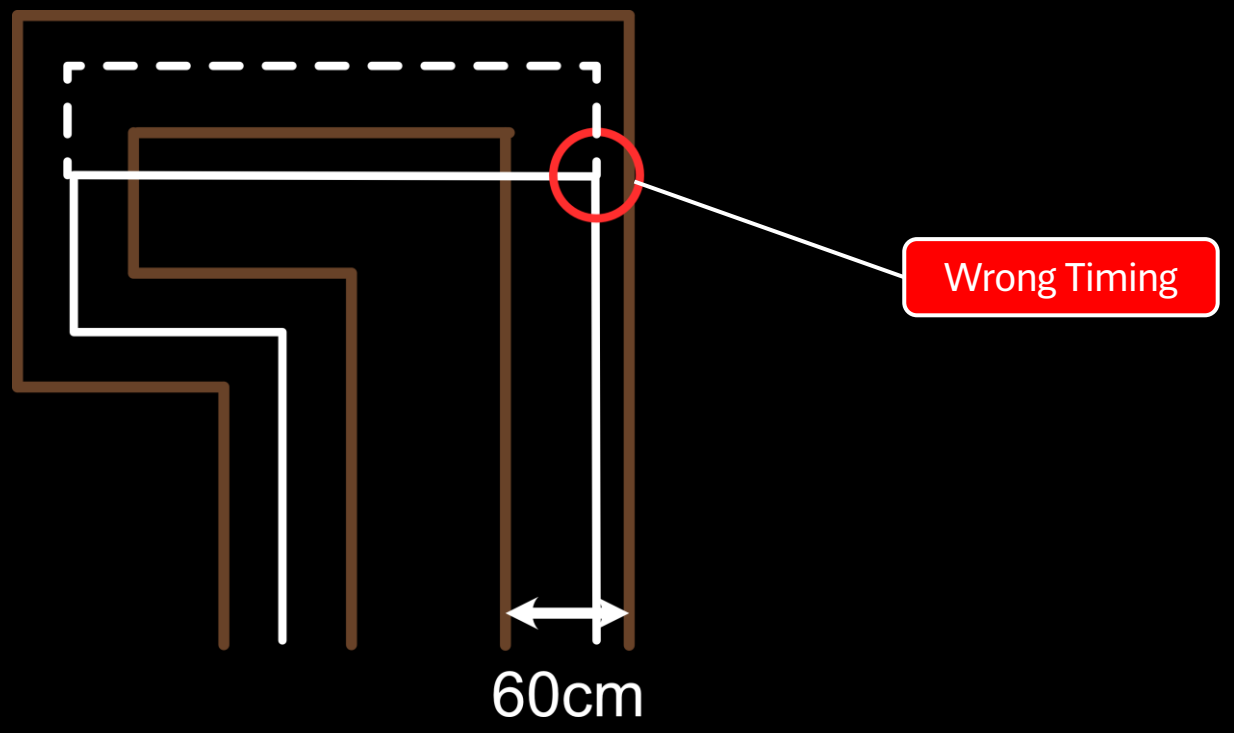
Test 1 **Test 2** **Test 3**

Users will unconsciously memorize the path

Wizard Errors

- 90% of all errors are due to wizard timing

➔ If we were to automate this, we suspect there would be fewer errors



Environment

VS

Cognitive Load?

Comparison with benchmark

Tactile Compass(Liu et al.)	SensePath (Develop 1)
Easy Path ➔ Better scores	More difficult
OptiTrack (mm-precisision) ➔ More precisision results	Video+Observation
No Stick ➔ Less cognitive load	Stick
Time in zone ➔ More accurate results	Steps in zone
Automatic feedback ➔ No human errors	Wizard of Oss

References

- **Liu, G., Yu, T., Yu, C., Xu, H., Xu, S., Yang, C., Wang, F., Mi, H., & Shi, Y. (2021).** Tactile Compass: Enabling visually impaired people to follow a path with continuous directional feedback. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. ACM. https://doi.org/10.1145/3411764.3445644
- **Anthropic. (2025).** Claude (Claude Opus 4.6) [Groot taalmodel]. https://claude.ai
- **OpenAI. (2025).** ChatGPT (GPT-4o) [Groot taalmodel]. https://chat.openai.com
- **Google DeepMind. (2025). Gemini** [Groot taalmodel]. https://gemini.google.com
- **Google DeepMind. (2025). Nano Banana Pro** (Gemini 3 Pro Image) [AI-beeldgeneratiemodel]. https://blog.google/innovation-and-ai/products/nano-banana-pro/

"All content in this document has been reviewed and approved by the author. AI tools were used solely to support text processing, coding, and image editing, not for generating research data or conclusions."